

Algorithms Problem Solving

Level 1

Rectangle Area



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Definition

- ◆ This program reads the length and width of a rectangle and calculates its area.
- ◆ The program uses functions, reference parameters, multiplication, and return values.

Explanation

- ◆ The `readNumbers()` function reads the rectangle's length and width.
- ◆ The parameters `int& length` and `int& width` are reference parameters, so the function stores the input directly in the variables from `main()`.
- ◆ The `RectangleArea()` function calculates the area by multiplying the length by the width.
- ◆ The function returns the calculated area as an `int`.
- ◆ The `printRectangleArea()` function displays the result and calls `RectangleArea()` to calculate the area.
- ◆ In `main()`, the functions are called in order:
 - ◆ `readNumbers(A, B);`
 - ◆ `printRectangleArea(A, B);`

Steps in main()

1. Create two integer variables for length and width.
2. Read the length and width.
3. Pass the values to RectangleArea().
4. Multiply length by width.
5. Return the calculated area.
6. Print the rectangle area.

💡 Example

◆ If the user enters:

Length: 10

Width: 5

◆ The area is calculated by multiplying 10 by 5.

➡ Result

Rectangle Area = 50

Note

- ◆ The area of a rectangle is calculated by multiplying its length by its width.
- ◆ Since the program uses int, the length, width, and result are whole numbers.

⊘ Warning

◆ If the length or width is negative, the program will still perform the multiplication. The code does not validate whether the dimensions are positive.



Comparison

Larger Rectangle

- ✓ Has a larger length, width, or both.
- ✓ Produces a larger area when the other dimension remains the same.
- ✓ Area depends on both dimensions.

Smaller Rectangle

- ✓ Has a smaller length, width, or both.
- ✓ Produces a smaller area when the other dimension remains the same.
- ✓ Area depends on both dimensions.

✦ Summary

- ✓ Rectangle area is calculated by multiplying length by width.
- ✓ Reference parameters allow readNumbers() to modify variables from main().
- ✓ A function can return a calculated value.
- ✓ One function can call another function and use its returned value.
- ✓ Separating input, calculation, and output makes the program easier to understand.

Thank you 😊